Insurance Policy Q&A RAG Chatbot

Complete Project Documentation

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1. What Is This Project?

This project is an **AI-powered chatbot that reads insurance policy documents and answers customer questions in plain English**.

A customer has a 90-page health insurance policy document. They want to know one thing — "Is my knee surgery covered?" Normally they would call the customer care line, wait 20 minutes, speak to an agent who reads the same document, and get an answer in 5 minutes. With this chatbot — they type the question, and the answer comes back in 3 seconds, with the exact page and clause from the policy that supports it.

The chatbot does not make up answers. It reads only from the actual policy documents uploaded into the system. Every answer includes a citation — which document, which section, which clause. This is what makes it trustworthy enough for insurance, healthcare, legal, and financial use.

This type of system is called a **RAG Chatbot — Retrieval Augmented Generation**.

The insurance domain is the focus of this project. But before going deep into insurance, it is important to understand that RAG is not just for insurance. Any industry that has large internal documents and customers or employees who need answers from those documents can use this exact same approach. The only thing that changes is the documents loaded into the system.

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2. Which Industries Can Use a RAG Chatbot?

RAG chatbots are one of the most widely applicable AI products being built right now. Every company in every industry has the same fundamental problem — they have important information locked inside documents, and people cannot find it fast enough.

Here is a quick overview of the major domains where RAG chatbots are being deployed today:

HR Policy Chatbot

The problem: Every company has an employee handbook, leave policy, travel reimbursement policy, promotion guidelines, and code of conduct. These documents sit on an internal portal that nobody reads. Employees ask HR the same questions every week — "How many sick leaves do I get?", "Can I carry forward earned leaves?", "What is the notice period for resignation?" HR teams spend 30 to 40% of their time answering questions that are already answered in documents.

What the RAG chatbot does: Employees ask on Slack, Teams, or a web portal. The chatbot reads the HR policy documents and answers instantly. HR focuses on real problems instead of repeat questions.

Who uses it: TCS, Infosys, Wipro, and any company with more than 500 employees. HR SaaS platforms like Darwinbox, Keka, and Zoho People are building this into their products.

Documents loaded: Employee handbook, leave policy, compensation structure, promotion framework, travel and expense policy, POSH policy, exit process document.

Legal Contract Chatbot

The problem: Lawyers and legal teams spend 60% of their time reading contracts — vendor agreements, client contracts, NDAs, lease deeds, employment contracts. A junior lawyer joining a firm must read 200-page agreements before they can answer a client question. Law firms charge by the hour — time reading documents is expensive.

What the RAG chatbot does: A lawyer uploads a contract and asks — "What is the penalty clause if we miss the delivery deadline?" or "Does this NDA cover subsidiary companies?" The chatbot finds the exact clause and answers in seconds.

Who uses it: Law firms (Cyril Amarchand, AZB & Partners), corporate legal departments, real estate companies, procurement teams, and contract management platforms like Ironclad and Agiloft.

Documents loaded: Vendor contracts, customer agreements, NDAs, lease agreements, employment contracts, regulatory filings, court judgements.

Medical and Clinical Chatbot

The problem: Hospitals and clinics generate enormous amounts of clinical documents — patient discharge summaries, lab reports, drug prescription histories, clinical trial protocols, medical guidelines. A doctor seeing 50 patients a day cannot read each patient's full history. A nurse needs to know the drug interactions for a patient on 6 medications.

What the RAG chatbot does: Doctor asks — "What was this patient's creatinine level trend over the last 6 months?" The chatbot reads the patient's lab reports and answers. A pharmacist asks — "Can Metformin and Lisinopril be given together to a 72-year-old diabetic with kidney disease?" The chatbot reads drug interaction guidelines and answers.

Who uses it: Apollo Hospitals, Fortis, Manipal Health, pharma companies, clinical research organisations. Also being built into EHR (Electronic Health Record) systems like Epic.

Documents loaded: Patient records, lab reports, clinical guidelines, drug interaction databases, treatment protocols, discharge summaries, insurance pre-auth forms.

Financial Report Chatbot

The problem: Equity research analysts, investment bankers, and fund managers read hundreds of pages of annual reports, quarterly earnings documents, and DRHP filings every week. A company's annual report is 300 pages. An analyst covering 20 companies reads 6,000 pages a quarter. They need specific numbers fast.

What the RAG chatbot does: Analyst asks — "What was Reliance Industries' EBITDA margin for the petrochemical segment in FY23 vs FY22?" The chatbot reads the annual reports and returns the exact figures with the page reference. "What are the top 5 risk factors in Zomato's latest annual report?" — answered in 10 seconds instead of 45 minutes of reading.

Who uses it: Mutual fund companies, brokerage firms, investment banks, PE funds, and financial research platforms like Bloomberg, Refinitiv, and Screener.in.

Documents loaded: Annual reports, quarterly results, DRHP filings, credit rating reports, RBI policy documents, SEBI circulars, analyst reports.

Compliance and Audit Chatbot

The problem: Companies in banking, insurance, and financial services must comply with hundreds of RBI, SEBI, and IRDAI regulations. New circulars come out every month. Compliance teams must read every circular and assess the impact on the company's operations. Missing a regulation can result in fines of crores.

What the RAG chatbot does: Compliance officer asks — "What does RBI's circular on digital lending from September 2022 say about cooling off periods?" The chatbot reads the circular and answers with the exact rule. An auditor asks — "According to SEBI's insider trading regulations, how many days before a board meeting must trading be frozen?" — answered instantly from the document.

Who uses it: Banks (HDFC, ICICI, Axis), NBFCs, insurance companies, stock brokers, and Big 4 audit firms (Deloitte, PwC, EY, KPMG).

Documents loaded: RBI Master Directions, SEBI regulations, IRDAI guidelines, Income Tax Act sections, Companies Act clauses, internal audit checklists, risk frameworks.

Education and EdTech Chatbot

The problem: Students have textbooks, lecture notes, study materials, and past exam papers — but finding the right answer at the right time is hard. A student preparing for UPSC, CA exams, or engineering entrance tests needs to search through hundreds of pages of notes quickly.

What the RAG chatbot does: Student asks — "Explain the difference between fiscal deficit and revenue deficit with an example from the 2023 Union Budget." The chatbot reads the course material and the budget document and explains it clearly.

Who uses it: Unacademy, Byju's, Vedantu, universities, coaching institutes, and UPSC preparation platforms.

Documents loaded: Textbooks, lecture notes, previous year question papers, syllabus documents, NCERT books, reference materials.

E-commerce and Retail Chatbot

The problem: E-commerce companies have thousands of product descriptions, return policies, shipping policies, seller guidelines, and warranty documents. Support teams answer the same questions manually thousands of times a day — "Can I return this after 15 days?", "What is the warranty for this refrigerator?"

What the RAG chatbot does: Customer asks — "I bought a Samsung washing machine 10 days ago. The spin cycle is not working. Is this covered under warranty and how do I raise a service request?" The chatbot reads the warranty document and the return policy and answers with the exact process.

Who uses it: Flipkart, Amazon India, Myntra, Meesho, and any retail brand with a large product catalog.

Documents loaded: Product descriptions, warranty documents, return and refund policy, shipping policy, seller handbook, FAQ database.

Government and Public Services Chatbot

The problem: Government portals have thousands of pages of scheme documents, eligibility criteria, application procedures, and guidelines. Citizens spend hours calling helplines or visiting government offices to get answers that are already written in documents.

What the RAG chatbot does: Citizen asks — "I am a marginal farmer with 1.5 acres of land in Karnataka. Am I eligible for PM Kisan Samman Nidhi?" The chatbot reads the scheme eligibility document and answers clearly. "What is the penalty for late GST filing for a company with turnover below ₹1.5 crore?"

— answered from the GST Act document.

Who uses it: DigiLocker, MyGov, state government portals, GST Network, and central government ministry helplines.

Documents loaded: Government scheme documents, eligibility guidelines, application procedures, tax circulars, state government GOs, RTI response templates.

Summary Table

Domain	Core Documents	Who Benefits	Key Questions
Insurance	Policy PDFs, claims guidelines	Policyholders, agents	Coverage, claims, benefits
HR	Employee handbook, leave policy	All employees	Leave, reimbursement, policy rules
Legal	Contracts, NDAs, agreements	Lawyers, legal teams	Clause lookup, obligations, risks
Medical	Patient records, drug guides	Doctors, nurses, admins	History, drug interactions, procedures
Finance	Annual reports, DRHP, circulars	Analysts, fund managers	Numbers, risks, year-on-year comparisons
Compliance	RBI/SEBI/IRDAI circulars	Compliance officers, auditors	Rules, timelines, penalties
Education	Textbooks, notes, past papers	Students, teachers	Concepts, exam topics, explanations
E-commerce	Product docs, warranty, returns	Customers, support agents	Eligibility, process, warranty
Government	Scheme docs, tax rules, GOs	Citizens	Eligibility, procedure, deadlines

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Why we chose Insurance for this project:
Insurance was chosen because the documents are long and complex, the stakes are high (wrong answers have real financial consequences), IRDAI compliance makes guardrails a mandatory requirement, and the volume of customer queries is enormous. This makes it the most challenging and most impressive domain to demonstrate.

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3. What Is RAG? — Explained Simply

RAG stands for **Retrieval Augmented Generation**. It is a technique that combines two things — searching through documents and using AI to write an answer based on what was found.

The best way to understand it is with an analogy.

Think of it like an open-book exam.

In a closed-book exam, you answer from memory. If you did not study a topic, you either guess or leave it blank.

In an open-book exam, you answer from the book. You search through the textbook, find the relevant page, read it, and write your answer based on what the book says. You are not guessing — you are writing from the source.

RAG works exactly like an open-book exam, but with AI and at the speed of milliseconds.

Step 1 — Retrieval (the searching part):
Before answering, the system searches through all the uploaded documents and finds the most relevant paragraphs for the user's question. These paragraphs are pulled out.

Step 2 — Generation (the answering part):
The AI model reads those retrieved paragraphs and writes the answer based on them. It answers only from what was found in the documents — not from general internet knowledge.

Why this matters:

- The answer always comes from the actual document — not from memory or guessing
- The model cannot invent policy clauses that do not exist
- Every answer includes a citation — "This is from Section 4.2, Page 23 of your policy"
- If the answer is not in the document, the system says so honestly — it does not make something up

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4. Why RAG? — What LLMs Do Not Know

This is the most important question. Why can we not just use ChatGPT or Claude directly? Why do we need RAG at all?

The answer is simple: **LLMs are trained on public internet data. Your company's documents were never on the internet.**

What LLMs Are Trained On

When OpenAI trained GPT-4, they used billions of pages from the public internet — Wikipedia, news articles, books, research papers, public websites, and forums. The training data was collected up to a certain date and then frozen.

This means GPT-4 knows:

- What health insurance is in general
- What terms like "pre-existing condition" or "cashless hospitalization" mean in general
- General facts about how insurance works worldwide

But GPT-4 does NOT know:

- What YOUR specific company's policy says
- What the exact sub-limit for cataract surgery is in Star Health's Comprehensive Plan 2024
- What the waiting period is in Bajaj Allianz's specific critical illness rider
- What changed in your policy when you updated it last month
- What your internal claims processing rules say
- What a specific customer's active policy covers

A Concrete Example

Ask ChatGPT: *"What is the room rent limit in Star Health Comprehensive Individual Plan for a sum insured of ₹5 lakhs?"*

ChatGPT will either say it does not have this specific information — or worse, it will give you a number that sounds correct but is completely made up. This is called **hallucination**. In insurance, a hallucinated answer can cause a customer to make a wrong financial decision. The company can face complaints, legal action, and regulatory penalties.

Now ask the same question to a RAG chatbot that has Star Health's policy document loaded. The system retrieves the exact clause and answers:

"For a ₹5 lakh sum insured under the Star Comprehensive plan, room rent is covered up to 1% of the sum insured per day — which means up to ₹5,000 per day. ICU charges are covered up to 2% per day, which is ₹10,000 per day."
[Source: Section 3.2 — Room and Board Expenses, Page 14, Star Comprehensive Policy Document 2024]

One is a guess. The other is pulled directly from the actual document. The difference is critical.

Three Reasons Why LLMs Cannot Replace RAG

Reason 1 — Training cutoff:
LLMs have a training cutoff date. If Star Health updated their policy in March 2024 and added a new maternity benefit, GPT-4 (trained before that) does not know about it. Your internal documents are always current. RAG uses your current documents, not outdated training data.

Reason 2 — Private data never reached LLMs:
Insurance policy documents, internal claims guidelines, agent manuals, customer Q&A logs — none of this was ever made public on the internet. LLMs have never seen it. They cannot answer questions about it no matter how advanced they are. The only way to use this data with AI is through RAG.

Reason 3 — Specificity:
General knowledge is not enough in insurance. A customer does not want to know what co-payment means in general. They want to know what co-payment applies to THEIR specific policy with THEIR sum insured. LLMs give general answers. RAG gives specific answers pulled from the exact document the customer signed.

The Rule to Remember

If the information exists inside your company and was never made public — LLMs do not know it. RAG is the only way to make AI answer questions about it accurately.

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5. Why This Data Is Internal Data — And Why That Matters

People sometimes ask — why can we not just Google the answers to insurance questions? The answer is that the most important data in this project is **internal data** — information that lives inside the company and is never available to the public.

What Is Internal Data?

Internal data is any information a company creates for its own operations and keeps private. It is not on the internet. It is not searchable by Google. It exists only inside the company's systems — on their servers, in their file storage, in their databases.

For an insurance company, this includes:

Policy documents: The general concept of health insurance is public knowledge. But the exact wording of HDFC Ergo's Policy Number HE-HEALTH-2024-COMP — with its specific exclusions, specific sub-limits, and specific definitions — that document is internal. HDFC Ergo wrote it, owns it, and it is not on Google.

Claims guidelines: How the claims department decides to approve or reject a claim is internal. The thresholds, document checklists, escalation rules — all internal. No customer can see this.

Customer Q&A history: When a customer called and the agent answered — that conversation record is stored in the company's CRM. It belongs to the company. It is never public.

Agent training manuals: How to handle difficult situations, what to say, what to check — this is proprietary knowledge built over years. Internal.

Product pricing logic: Why one customer pays ₹8,000 and another pays ₹14,000 for the same plan — the rules behind this are internal. This is a competitive secret.

Why You Cannot Send Internal Data to Public ChatGPT

This concern comes up in every client meeting. The Compliance Officer will ask: *"If we use ChatGPT, does our policy data go to OpenAI's servers?"*

The concern is valid. If you send a policy document to ChatGPT's API, that content travels to OpenAI's servers in the USA. For insurance companies, this creates three problems:

Problem 1 — Data Privacy: Policy documents may contain pricing information and proprietary business logic that competitors would love to see.

Problem 2 — Regulatory Compliance: IRDAI requires insurers to maintain data security. Sending policy data to a third-party server outside India may violate data localisation requirements. RBI has similar rules for financial data.

Problem 3 — Customer Data Risk: If any customer information is in the documents — even indirectly — this creates DPDP Act (India's Digital Personal Data Protection Act, 2023) compliance issues.

How RAG Solves This

RAG keeps internal data inside your own control.

Policy documents are stored in your own AWS S3 bucket — inside your own AWS account, in the India region. The embeddings are stored in your own RDS database. When a user asks a question, only a few retrieved paragraphs are sent to the LLM to generate the answer — not the full document.

With AWS Bedrock, you can use Claude or GPT-4 without your data ever leaving AWS infrastructure. AWS guarantees that data is not used for training their models.

For the highest security requirements — government insurers or large banks — the LLM itself can be hosted on-premise using open source models like LLaMA 3. In this setup, not a single byte of internal data leaves the company's servers.

The RAG architecture is specifically designed to let you use AI on internal, confidential data — safely and in compliance with regulations.

What Happens When This Is Not Handled Correctly

Here is what goes wrong in the real world: A bank employee copies an internal credit policy document and pastes it into free ChatGPT to ask questions. That document — with internal loan pricing rules — is now on OpenAI's servers. The bank has no idea this happened. Nobody approved it.

This is called **shadow AI** — employees using public AI tools with company data without authorisation. It is happening right now in thousands of companies. A properly deployed internal RAG system is the enterprise-grade alternative that keeps data secure.

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6. Why Do We Need This Chatbot in Insurance?

India's insurance industry handles over **50 crore policies** across health, motor, life, and general insurance. Every single one of those policies is a document — ranging from 40 pages to 120 pages — written in dense legal language that most people cannot understand.

Here is what happens today without this system:

Customer side:

- Customer receives a 90-page policy PDF by email
- They do not read it — too long, too complex
- When they need to make a claim, they do not know what is covered
- They call the helpline and wait — average wait time in India is 8 to 15 minutes
- The agent reads the same document and tells them the answer
- If the agent gives wrong information, the customer gets a claim rejected later
- Customer is angry, files a complaint, sometimes goes to the insurance ombudsman

Company side:

- HDFC Ergo alone handles over 1.2 lakh calls per day across all products
- 65 to 70% of these calls are about policy coverage questions
- These are questions already answered in the policy document
- Each call costs the company approximately ₹80 to ₹150 in agent time
- That is ₹60 lakh to ₹1.3 crore spent every single day answering repeat questions
- Agent turnover is high — training new agents is expensive and takes months

The RAG chatbot eliminates the majority of these calls. Customers get instant answers. Agents handle only complex escalations. Company saves crores. Customer satisfaction improves.

Why Now?

Three things came together in 2023-2024 that make this possible right now:

Large Language Models became good enough. GPT-4, Claude, and Gemini can read complex legal text and explain it in simple language. Earlier AI systems could not do this reliably.

Vector databases became production-ready. pgvector, Pinecone, and Weaviate can store millions of document chunks and find the relevant ones in milliseconds at scale.

IRDAI's Digital Push. The Insurance Regulatory and Development Authority of India has been pushing insurers to improve digital customer service since 2022. Companies that deploy AI chatbots now are ahead of the regulation curve.

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7. The Five Real Problems This Solves

Problem 1 — Policy Documents Are Unreadable

Here is an actual sample of how insurance policies are written:

"The indemnification shall be limited to expenses reasonably and necessarily incurred for treatment of illness or accidental bodily injury sustained during the policy period, subject to the sub-limits specified in the Schedule of Benefits annexed hereto, excluding pre-existing conditions as defined under Section 2(m) of this policy."

No normal person understands this on first read. The chatbot translates it to: *"Yes, your surgery is covered — but only up to the amount listed in your benefits table. Pre-existing conditions from before your policy start date are not covered."*

Problem 2 — Wrong Claims Get Filed

When customers do not understand their policy, they file claims for things that are not covered. This wastes their time and the company's time. The chatbot prevents this by telling the customer upfront — *"This treatment is not covered under your plan. Here is what IS covered for this condition."*

Problem 3 — Right Claims Get Missed

The opposite also happens. Customers do not claim for things they are actually entitled to — because they never knew the benefit existed. Add-on covers, no-claim bonuses, free health check-ups, wellness benefits — these go unclaimed because the customer never read page 67 of their policy. The chatbot surfaces these proactively: *"You are also entitled to a free annual health check-up. Would you like to know how to book it?"*

Problem 4 — Renewal Decisions Made Without Information

At renewal time, customers decide whether to continue, switch, or upgrade their policy. Most make this decision without understanding what they currently have versus what alternatives offer. The chatbot makes this comparison easy — questions answered in real time during the renewal conversation.

Problem 5 — Agent Knowledge Is Inconsistent

Different call centre agents give different answers to the same question depending on their experience and how recently they were trained. The chatbot gives the same correct answer every time, pulled directly from the official policy document. IRDAI mandates that customers receive accurate information — consistency is a compliance requirement, not just a nice-to-have.

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8. Who Are the Customers?

This is a B2B product. The buyers are companies that deal with large volumes of policy documents and customer queries.

Customer 1 — Health Insurance Companies

Companies: Star Health Insurance, Niva Bupa, Care Health, HDFC Ergo Health, Bajaj Allianz Health

Their exact problem:
Star Health has over 500 different health plan variants — individual plans, family floater plans, senior citizen plans, critical illness plans, maternity plans, diabetic plans, and more. Each plan has different coverage limits, different exclusions, different co-payment rules, and different sub-limits for specific treatments. Even their own agents find it difficult to remember all the differences.

How they use the chatbot:

- Customer portal — policyholder logs in and asks questions about their specific active policy
- Agent assist tool — call centre agent uses the chatbot to get answers quickly during a live call
- WhatsApp chatbot — customer sends a message on WhatsApp and gets an instant reply
- Renewal assistant — chatbot helps customers understand what they are renewing before they pay

Specific questions customers ask:

- "Is LASIK eye surgery covered in my plan?"
- "What is the waiting period for maternity benefits?"
- "My father is 68 years old — can he be added to my family floater?"
- "What is the maximum room rent my plan covers per day?"
- "Is Ayurveda or homeopathy treatment covered?"
- "I was diagnosed with diabetes 2 years ago — will my new policy cover it?"

Business impact:
A health insurer with 10 lakh active policyholders receives approximately 25,000 queries per day. If the chatbot handles 70% automatically, the company saves 17,500 human interactions per day. At ₹100 per interaction, that is ₹17.5 lakh saved every single day.

Customer 2 — Motor Insurance Companies

Companies: HDFC Ergo Motor, Bajaj Allianz Motor, ICICI Lombard, Acko, Digit Insurance, Tata AIG

Their exact problem:
Motor insurance customers are most active at two moments — when they buy the policy and when they have an accident. At the accident moment, they are stressed, sometimes injured, and need immediate clear answers. They do not have time to read a policy document.

How they use the chatbot:

- Accident support chatbot — triggered immediately when customer is in distress
- Policy comparison tool — during new purchase, customer asks which plan suits their needs
- Claim status chatbot — customer tracks their claim and understands what documents are needed
- Renewal assistant — explains no-claim bonus calculations and add-on recommendations

Specific questions customers ask:

- "My car was flooded in the rain — is water damage covered?"
- "I hit a stray animal and my bumper is damaged — who pays?"
- "What is the IDV of my car right now?"
- "I have a zero depreciation add-on — does that mean I get full repair cost?"
- "My car was stolen — what documents do I need to file the claim?"
- "The other driver was at fault — can I claim from their insurance?"

Customer 3 — Life Insurance Companies

Companies: LIC, HDFC Life, ICICI Prudential Life, SBI Life, Bajaj Allianz Life, Max Life

Their exact problem:
Life insurance policies are long-term — 10, 20, or 30 year contracts. Over time, customers forget what they signed up for. They have questions at different life stages — when they get married, have children, change jobs, or approach the maturity date.

How they use the chatbot:

- Policy servicing chatbot — handle routine requests without a human agent
- Nominee update assistant — guide customer through the process step by step
- Maturity benefit explainer — tell customer what they will receive and when
- Surrender value tool — tell customer what they get if they stop paying premiums early

Specific questions customers ask:

- "What happens to my policy if I miss 3 premium payments?"
- "How do I change my nominee from my father to my wife?"
- "I took a policy loan 2 years ago — how much do I still owe?"
- "If I die in a road accident, does my family get more than the basic sum assured?"
- "What is the surrender value of my policy today?"
- "Can I pause my premiums for 6 months if I lose my job?"

Customer 4 — General Insurance Companies

Companies: New India Assurance, Oriental Insurance, United India, Bajaj Allianz General, Reliance General

Their exact problem:
General insurance covers everything from home insurance to travel insurance to marine cargo to fire insurance. Each type has completely different terms. The same company may offer 50 different policy types. Managing customer queries across all of them with human agents is expensive and error-prone.

How they use the chatbot:

- Travel insurance chatbot — customer asks what is covered during an international trip
- Home insurance assistant — explains what damage types are and are not covered
- SME business insurance chatbot — small business owners understand their commercial policy

Specific questions customers ask:

- "My laptop was stolen from my hotel room during a business trip — is it covered?"
- "I am going to the US for 3 weeks — does my plan cover medical emergencies?"
- "My house flooded — what is the claim process and how long does it take?"
- "My shop had a fire — does my commercial policy cover the stock that was destroyed?"

Customer 5 — Insurance Brokers and Aggregators

Companies: Policybazaar, Coverfox, Turtlemint, Insurance Dekho, Ditto Insurance

Their exact problem:
Insurance aggregators show policies from 20 to 30 different companies on the same platform. A customer wants to compare a Star Health plan versus a Niva Bupa plan. Without the chatbot, they must read two separate 80-page documents and compare them manually — which nobody does.

How they use the chatbot:

- Policy comparison chatbot — customer asks one question and gets the answer from both policies side by side
- Recommendation engine — based on customer's questions, suggest the most suitable plan
- Pre-purchase assistant — help customers understand what they are buying before they pay

Specific questions customers ask:

- "Between Star Health and Niva Bupa, which one covers OPD consultations?"
- "Which plan has a shorter waiting period for pre-existing conditions?"
- "I am 45 years old with a history of hypertension — which plan is right for me?"

Customer 6 — Corporate HR Departments

Companies: Any large company offering group health insurance to employees — TCS, Infosys, Wipro, Reliance, HDFC Bank, and thousands of others

Their exact problem:
When a company offers group health insurance, one HR team manages thousands of employees. Every time an employee has a medical question, they message HR. HR does not know every detail of the insurance policy either. The HR team spends hours every week answering the same insurance questions.

How they use the chatbot:

- Employee self-service portal — employees get answers about their group health policy without involving HR
- Onboarding assistant — new employees understand their benefits from day one
- Claims guidance tool — step-by-step help when an employee needs to file a claim

Specific questions employees ask:

- "Does my insurance cover my parents who live with me?"

- "What is the maternity benefit limit for my plan?"
- "I need a cataract surgery for my mother — is she covered?"
- "Which hospitals near my home are part of the cashless network?"

Customer 7 — Third Party Administrator (TPA) Companies

Companies: Medi Assist, Vidal Health TPA, Paramount Health Services, Health India TPA

Their exact problem:
TPAs sit between insurance companies and hospitals. They process cashless claims — when a patient is admitted, the TPA verifies coverage and approves the cashless facility. TPA agents need to quickly check policy details during active hospital admissions — sometimes at midnight during emergencies.

How they use the chatbot:

- Agent assist tool — TPA staff quickly verify coverage during cashless admission
- Pre-authorization assistant — check if a procedure is covered before approving
- Document checklist generator — tell the hospital exactly what documents are needed

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9. How a Real Client Discussion Happens

This section shows how a project like this actually begins — from the first call to the signed contract. Students never see this part in courses. This is the real world.

Stage 1 — The First Call

The first contact is usually informal — a LinkedIn message, a referral, or a conference. The client does not start with a detailed brief. They start with a pain point.

A typical first conversation with a health insurance company:

Client (Head of Digital, Star Health): "We are getting flooded with calls. 70% of the questions our agents answer every day are about policy coverage. The same 50 questions, over and over. We want to do something with AI. Can you help?"

Data Scientist: "Yes. How many policies do you have active right now?"

Client: "Around 30 lakh policyholders. We have about 200 different plan variants."

Data Scientist: "And when a customer calls with a coverage question, what do your agents do? Do they have a system or do they look at the policy document directly?"

Client: "They have a knowledge base but it is outdated. Most of them just search the PDF."

Data Scientist: "So the information is already in the documents — the problem is speed of access and consistency of answers?"

Client: "Exactly. And we cannot afford to give wrong information. If an agent wrongly tells a customer their surgery is covered and it is not — that is a complaint, sometimes a legal issue."

Data Scientist: "We can build a system where your policy PDFs are the single source of truth. The AI reads only from those documents. Every answer comes with the exact section of the policy it came from."

This is the point where the client says — "Tell me more. Can you come in for a proper meeting?"

Stage 2 — The Requirements Meeting

A week later, the team sits down — typically 4 to 6 people on the client side:

- Chief Digital Officer or Head of Digital Transformation
- Head of Customer Service
- Head of IT or CTO
- A Compliance Officer (very important in insurance)
- A Product Manager

On the data science side:

- Project Lead
- Senior AI Engineer
- Solutions Architect

The conversation gets very specific.

Client (Compliance Officer): "Every answer this chatbot gives — we need to know which document it came from and which exact section. IRDAI requires us to maintain audit logs of all customer interactions."

Data Scientist: "Every response will include the source — document name, section number, page number. All interactions are logged with timestamp, user ID, and the full conversation. Nothing is deleted."

Client (Head of Customer Service): "What happens when the chatbot does not know the answer? We cannot have it making things up."

Data Scientist: "If the answer is not in the uploaded documents, the chatbot says — I cannot find this information in your policy, please contact your advisor. It does not guess."

Client (Head of IT): "Can this connect to our CRM? When a customer asks a question, can the chatbot also pull their specific policy details — their sum insured, their renewal date?"

Data Scientist: "Yes — through an API integration. The chatbot can pull answers specific to their plan, not generic answers."

Client (Product Manager): "How long does it take to add a new policy? If we launch a new product next month, how quickly can it be in the chatbot?"

Data Scientist: "A new policy PDF can be uploaded and indexed in under 10 minutes. It is available to query immediately. No retraining needed."

Stage 3 — The Data Discussion

After the requirements meeting, the data science team asks very specific questions about what documents exist and in what format.

Data Scientist: "Can you share samples of the policy documents? We need to understand the format, language, and structure."

Client sends: 5 sample policy PDFs — one health plan, one motor plan, one term life plan, one critical illness rider, one travel plan.

Data Scientist reviews and notes:

- Most are PDFs with proper text layers — good, can be extracted directly
- Two are scanned documents — need OCR processing before they can be read
- Policy documents have tables for sub-limits and benefit amounts — need special handling
- Some have exclusions listed in very small footnotes — need to make sure these are captured

- One document has 11 annexures — each annexure is a separate coverage section

The data science team comes back with specific questions:

- "Do all your policies exist as PDFs, or are some in HTML on your website?"
- "Are the scanned documents all pre-2018? Will future policies always be digital?"
- "The benefit tables — are these in the PDF or do they come from a separate system?"
- "Do you have FAQs or previous agent answers? This helps us test the system."
- "Do you have transcripts from past customer calls? This is valuable for testing accuracy."

Stage 4 — Proof of Concept

Before a full contract is signed, clients almost always want a proof of concept — a small working demo built on 5 to 10 of their actual documents.

The data science team takes 5 real policy PDFs, builds a basic RAG system in 2 weeks, and demonstrates it live. The demo covers:

- Customer asks a coverage question — gets an answer with citation
- Customer asks something not in the policy — system says it does not know
- Agent uses the tool during a simulated customer call
- Compliance officer sees the audit log

If the demo works, the contract is signed. This is how most AI projects in India start today.

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10. What Data Do We Use?

The quality and variety of data determines whether the chatbot answers well or poorly. For this project, we use **7 different types of data**.

Data Type 1 — Policy Documents (Primary Source)

This is the core of the entire system. Everything else supports this.

What it is:
The actual insurance policy documents — the legal contracts between the insurance company and the policyholder.

What they contain:

- Policy schedule — the customer's name, policy number, sum insured, premium amount, policy period
- Coverage section — what is covered, under what conditions, and up to what limit
- Exclusions section — what is specifically NOT covered
- Definitions section — what terms like "hospitalization", "pre-existing disease", "cashless" actually mean in legal language
- Claims procedure — step by step how to file a claim
- Conditions and warranties — rules the customer must follow to keep the policy valid
- Annexures and riders — additional coverage added on top of the base policy
- Sub-limits table — specific limits per treatment type (e.g., cataract surgery maximum ₹40,000)
- Network hospital list — which hospitals offer cashless treatment
- Grievance redressal section — how to complain if things go wrong

Format: PDF (most common), HTML (for web-based policies), Word documents (for draft policies)

Volume: A single insurer like Star Health has 200+ plan variants. Each plan has a main document plus annexures. Total document count could be 500 to 800 PDFs.

Challenges:

- Old policies are scanned images — OCR is needed to extract text
- Tables with benefit amounts need special extraction logic
- Legal language is dense — splitting a clause in the middle makes it meaningless
- The same term can mean different things across different policies

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Data Type 2 — Policy Wordings and Product Manuals

What it is:
Internal documents that insurance companies create for their agents — simpler explanations of what each policy covers, written in plain language for

training purposes.

What they contain:

- Feature summaries — "This plan covers X, Y, Z"
- Comparison tables — how this plan differs from competitor plans
- FAQs — common questions agents hear and the correct answers
- Sales scripts — how to explain benefits to customers
- Product update notices — what changed and from when

Why this is valuable:
These documents are already written in simpler language. When the chatbot retrieves from these alongside the official policy, the answers become clearer and easier to understand. The official policy provides legal accuracy. The product manual provides plain English explanation.

Data Type 3 — Claims Data and Claims Guidelines

What it is:
Documents that describe how claims are processed — what documents are required, what the timelines are, why claims get rejected, and how to appeal a rejection.

What they contain:

- Claim form templates with field-by-field explanations
- Document checklist for different claim types — surgery, accident, theft, travel emergency
- Claim processing timelines — "cashless claims approved in 2 hours, reimbursement in 15 days"
- Common rejection reasons — "claim filed after 30 days of discharge" or "treatment at non-network hospital"
- Appeals process — how to challenge a rejected claim
- TPA guidelines — instructions for hospitals on how to submit pre-authorization requests

Why this matters:
The most stressful moment for a customer is when a claim is rejected. If the chatbot can explain exactly why and what documents are needed to appeal, it reduces frustration significantly. This is one of the highest-value use cases.

Data Type 4 — Regulatory and Compliance Documents

What it is:
IRDAI circulars, guidelines, and regulations that govern how insurance companies must operate. These are public documents published by the Insurance Regulatory and Development Authority of India.

What they contain:

- Standard definitions all insurers must follow — for example, what counts as a "pre-existing disease"
- Mandatory benefits — IRDAI mandates that all health plans cover mental health treatment since 2022
- Waiting period rules — IRDAI caps the initial waiting period at 30 days for most conditions
- Portability guidelines — a customer's right to switch insurers without losing accumulated benefits
- Grievance handling timelines — insurer must respond to complaints within 14 days

Why this matters:
When a customer asks about their rights — "Can the insurance company reject my claim for a mental health hospitalization?" — the answer must be based not just on the policy document but also on what IRDAI mandates. The chatbot must know regulatory context.

Source: All IRDAI circulars are publicly available on irdai.gov.in. There are over 2,000 circulars published since 2000. For this project, we focus on circulars from 2018 onwards.

Data Type 5 — Customer Q&A History

What it is:
Historical records of questions customers have asked in the past — through call centre transcripts, email support tickets, chat logs, and WhatsApp messages.

What they contain:

- The question exactly as the customer asked it — natural language, often with spelling mistakes and mixed Hindi-English
- The answer the agent gave
- Whether the customer was satisfied

- Whether the case was escalated — which tells us when the agent could not answer
- Resolution time

Why this is the most valuable data for testing:
This data tells us exactly what real customers ask. We use it two ways:

First, as a test dataset — we take 500 real historical questions, run them through the chatbot, compare the chatbot's answer to the agent's answer, and measure accuracy.

Second, as a signal for improvement — questions the chatbot answers poorly tell us where we need better data or better chunking.

Privacy note: This data contains customer names and policy numbers. Before using it, all PII must be removed — names replaced, policy numbers masked. This is a mandatory pre-processing step.

Data Type 6 — Hospital and Network Data

What it is:
Information about hospitals that are part of the insurance company's cashless network.

What they contain:

- Hospital name, address, city, pincode
- Specializations available at each hospital
- Whether the hospital offers cashless admission for which plans
- Contact details for the hospital's insurance desk
- Pre-authorization process for that specific hospital
- Any special conditions or limits for that hospital

Why customers need this:
"Which hospitals near Koramangala, Bangalore accept cashless for my Star Health policy?" is one of the top 10 most asked questions. This data needs to be in the system and queryable by location.

Format: Excel sheets or CSV files. Updated quarterly. Some companies have real-time APIs.

Volume: A large health insurer like Star Health has 14,000+ network hospitals across India.

Data Type 7 — Product Comparison Data

What it is:
Side-by-side comparisons of different plans offered by the same insurer, or comparisons across multiple insurers.

What they contain:

- Feature comparison tables — does Plan A or Plan B cover OPD? Which has higher room rent?
- Premium comparison — what does each plan cost for different age groups and sum insured levels
- Waiting period comparison — Plan A has 2-year pre-existing waiting period, Plan B has 3 years
- Add-on availability — which riders are available on which base plans

Why this matters:
For insurance aggregators like Policybazaar, this is the primary use case. The chatbot answers comparison questions — "Between these two plans, which is better for a family of four with elderly parents?"

Source: Publicly available on company websites and aggregator platforms.

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11. Where Do We Get the Data?

Policy Documents:

- Directly from the insurance company — they share their document library
- Many insurers publish their policy wordings on their website and on IRDAI's portal
- IRDAI's policy document repository — some standard documents are publicly available

Claims Guidelines:

- From the insurer's internal document management system
- From TPA partners who have their own documentation

- From IRDAI guidelines that standardize certain claim processes

Regulatory Documents:

- IRDAI website (irdai.gov.in) — all circulars are public
- Insurance Institute of India publications
- Government of India finance ministry notifications

Customer Q&A History:

- CRM exports from the insurer — requires a data sharing agreement
- Call centre transcript exports — often in audio format, needs transcription first
- Email support ticket exports — usually available as CSV from helpdesk tools
- WhatsApp Business API message logs — if the insurer uses WhatsApp for support

Hospital Network Data:

- Direct from the insurer's provider management team
- From TPA databases — Medi Assist and Vidal Health maintain their own hospital databases
- From the NHA (National Health Authority) for Ayushman Bharat network hospitals

Product Comparison Data:

- Scraped from Policybazaar and other aggregator websites (publicly available)
- From the insurer's product team in structured Excel format
- From IRDAI's product filing database — all approved products are on record

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12. What Does the Data Look Like?

Let us look at actual examples to make this concrete.

Policy Document Sample — What It Says vs What the Customer Needs

Original text in the policy document:

"Hospitalization expenses shall be covered provided the Insured Person has been hospitalized for a continuous period of not less than twenty-four (24) hours, excluding Day Care Procedures listed in Annexure III, which shall be covered regardless of duration of hospitalization, subject to the condition that such Day Care Procedure is performed in a Hospital or Day Care Centre and necessitates specialized infrastructural requirements."

What the customer actually asks:

"Do I need to be admitted for a full day to claim insurance?"

What the chatbot answers:

"Yes, generally you need to be in the hospital for at least 24 hours to make a claim. But there is an important exception — certain procedures listed in Annexure III of your policy, like cataract surgery, do not require a 24-hour stay. These are called Day Care procedures. For these, even a few hours of admission is enough. The procedure must be done in a proper hospital or day care centre."

[Source: Section 2.1 — Hospitalization Definition, Page 8 of your policy]

This is the core transformation the system does — legal language to plain English, with the source cited.

Customer Q&A History Sample

This is what a record from the call centre database looks like after PII is removed:

```
Ticket ID: TKT-2024-0045231
Date: 14 March 2024
Channel: Inbound Call
Duration: 6 minutes 42 seconds
Plan: Star Comprehensive Individual – Sum Insured ₹5 Lakhs

Customer Question (transcribed):
"My doctor suggested I need a knee replacement surgery.
I have had knee problems for the last 5 years.
Will my insurance cover this? I took the policy 8 months ago."

Agent Answer:
"For joint replacement surgeries, there is a 2-year waiting period
```

```
from the policy start date for pre-existing conditions.
Since knee problems existed before you took the policy and your
policy is only 8 months old, this surgery would not be covered now.
You would need to wait until your policy completes 2 years."

Customer Satisfaction Rating: 3/5
Escalated: No
Resolution: Fully resolved
```

This kind of record tells us exactly how customers ask questions in real language — not in perfect English, with context about their situation — and what the correct answer looks like.

Hospital Network Data Sample

Hospital Name	City	Pincode	Specializations	Cashless Plans	Insurance Desk	Pre-Auth Time
Manipal Hospital	Bengaluru	560017	Cardiology, Ortho, Neuro	All Star Health Plans	080-25023456	2 hours
Apollo Hospitals	Chennai	600006	All specializations	All plans including OPD	044-28290200	4 hours planned, immediate emergency
Fortis Memorial	Gurugram	122002	Cardiac, Cancer, Transplant	Star Comprehensive, Star Family	0124-4921021	6 hours for elective surgery

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IRDAI Regulatory Data Sample

Circular: IRDAI/HLT/REG/CIR/166/09/2022
Subject: Inclusion of Mental Health Treatment in Health Insurance

All health insurance policies issued or renewed after September 2022 must cover mental health conditions on par with physical health conditions. Insurers cannot reject claims for psychiatric hospitalization, psychotherapy, or mental health medications.

How this enters the chatbot: When a customer asks "Does my policy cover depression treatment?" — the chatbot checks the policy document AND surfaces this IRDAI mandate if the policy was issued after September 2022, ensuring the customer knows their legal rights even if the policy wording does not explicitly state it.

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13. How the System Works — Step by Step

When a New Document Is Uploaded

1. Insurance company uploads a PDF policy document
2. System extracts all text from the PDF
3. If it is a scanned document, OCR is run first to convert image to text
4. The text is split into meaningful chunks — not by word count, but by meaning boundaries. A clause stays together. A table stays together.
5. Each chunk is converted into a numerical representation called an embedding — a list of numbers that captures the meaning of the text
6. All embeddings are stored in a vector database along with metadata — which document, which section, which page, which plan name
7. The document is now searchable. This whole process takes 2 to 5 minutes for a 90-page document.

When a Customer Asks a Question

1. Customer types — "Is my cataract surgery covered?"
2. System checks the input for safety — no prompt injection, no PII, question is about insurance
3. The question is also converted into an embedding
4. The system searches the vector database for chunks most similar in meaning to the question
5. Top 20 most relevant chunks are retrieved
6. A cross-encoder model re-reads the question and all 20 chunks together and re-ranks them — the top 5 after re-ranking are the most relevant
7. These 5 chunks are sent to the language model (GPT-4 or Claude) along with the question
8. The language model writes an answer in plain English — using only what the chunks say
9. A second check runs — does the answer actually come from the retrieved chunks? If not, it is flagged as low confidence
10. The answer is returned to the customer with the source citation — document name, section, page
11. The entire interaction is logged — timestamp, question, retrieved chunks, answer, confidence score

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14. What Users Ask — And How the Chatbot Responds

These are real examples based on actual call centre patterns from Indian insurance companies. Notice how informal and natural the questions are — customers type the way they speak.

Health Insurance — Coverage Questions

Q: "Is my knee replacement surgery covered?"

"Yes, knee replacement surgery is covered under your plan. However, there is a 2-year waiting period for joint replacement surgeries from your policy start date. Your policy started on 12 June 2023, so the waiting period will be over on 12 June 2025. If you have this surgery after that date, it will be fully covered up to your sum insured of ₹5 lakhs.

If the surgery is urgent and cannot wait, please speak with your doctor about whether any prior authorisation or exception applies."

[Source: Section 4.1 — Waiting Periods, Page 18 and Section 5.3 — Surgical Procedures, Page 27]

Q: "what about my parents? they are 65 and 68. are they covered in my plan?"

"Under your current Family Floater plan, your parents are listed as covered members. Your father (age 68) and mother (age 65) are both included in your policy schedule.

One important thing to note — for members above the age of 60, a co-payment of 20% applies. This means for any claim by your parents, you will need to pay 20% of the bill yourself, and the insurance covers the remaining 80%.

For example, if the hospital bill is ₹1,00,000, you pay ₹20,000 and the insurance pays ₹80,000."

[Source: Section 7 — Co-payment Clause, Page 34]

Q: "i was diagnosed with high BP 3 years back, will that be covered?"

"High blood pressure (hypertension) is a pre-existing condition since it was diagnosed before your policy start date. Under your plan, pre-existing conditions have a waiting period of 3 years from the policy start date.

Your policy started on 1 April 2022. This means hypertension and any conditions directly related to it will be covered from 1 April 2025 onwards — which has now passed.

So yes, hypertension treatment is now covered under your policy. If you are currently on medication for blood pressure, those costs may also be covered under the OPD benefit. Shall I check what your OPD limit is?"

[Source: Section 2 — Definitions: Pre-Existing Disease, Page 6 and Section 4.1 — Waiting Period Table, Page 18]

Q: "is mental health covered? like therapy sessions and all"

"Yes. Since 2022, IRDAI has made it mandatory for all health insurance policies to cover mental health conditions on the same basis as physical health. Your policy complies with this regulation.

What is covered:

- Psychiatric hospitalization — covered like any other hospitalization
- Outpatient psychiatric consultations — covered under your OPD benefit up to ₹5,000 per year
- Prescribed mental health medications — covered under the pharmacy benefit

What is not covered under your specific plan:

- General counselling or life coaching not prescribed by a psychiatrist
- Residential wellness programs or retreats

If you need to see a psychiatrist, you can go to any network hospital or empanelled clinic and use your OPD card."

[Source: IRDAI Circular IRDAI/HLT/REG/CIR/166/09/2022 and Section 6 — OPD Benefits, Page 31]

Claim Questions

Q: "I got admitted to a hospital last week and paid from my pocket. how do i get my money back?"

"You need to file a reimbursement claim. Here is the step-by-step process:

Step 1 — Collect all documents from the hospital:

- Original discharge summary
- All original bills and receipts (itemised bills)
- All diagnostic test reports (blood tests, X-rays, scans)
- Doctor's prescription for all medications purchased
- If surgery was done — the operation notes

Step 2 — Fill the claim form. Download it from the Star Health website. The form has two parts — one you fill, one the treating doctor fills.

Step 3 — Submit within 30 days of discharge. Your discharge was last week, so you still have time.

Step 4 — Track your claim. Reimbursement claims are processed within 15 working days. You will receive updates by SMS and email.

Shall I tell you the nearest branch where you can submit documents in person?"

[Source: Section 9 — Claims Procedure, Page 45]

Q: "my claim was rejected. it says pre existing condition. but i never had this disease before"

"I understand this is frustrating. A rejection is not final — you have the right to appeal.

Step 1 — Request the rejection letter in writing. It must state the exact reason with a reference to the policy clause.

Step 2 — Get a letter from your treating doctor clearly stating this condition was NOT pre-existing — it was not diagnosed or treated before your policy start date.

Step 3 — File a formal appeal with the Grievance team within 15 days of receiving the rejection. Email: grievance@starhealth.in. Quote your rejection reference number.

Step 4 — If the appeal is also rejected, escalate to the Insurance Ombudsman. This is a free service. The ombudsman is independent and can overturn the insurer's decision. Your nearest ombudsman office is in Bengaluru.

Your case sounds like it may be a documentation issue. The medical letter from your doctor is the most important piece of evidence. Would you like help understanding exactly what the rejection letter says?"

[Source: Section 10 — Grievance Redressal, Page 52 and IRDAI Ombudsman Guidelines 2017]

Motor Insurance Questions

Q: "it rained heavily yesterday and water entered my car engine. covered?"

"This depends on how the water entered.

If your car was stationary and water entered due to flooding — Yes, this is covered under your Comprehensive plan. It falls under Act of God or natural calamity coverage.

However — if you tried to start the car after seeing water inside, or drove through the flooded area, that is considered hydrostatic lock and may be treated as consequential damage caused by your own action. This is specifically excluded in most motor policies.

What you should do right now:

1. Do not try to start the car — this could make the damage worse and affect your claim
2. Call our 24-hour claims helpline and register a claim immediately
3. Take photos of the flooded area and your car before moving it
4. We will send a surveyor within 24 hours"

[Source: Section 3, Clause 3.4 — Natural Calamities, Page 12 and Section 4, Clause 4.7 — Consequential Damage, Page 21]

Q: "someone scratched my car in a parking lot and ran away. no witnesses. what do i do?"

"This is an own damage claim — you claim from your own insurance, not the other person's.

Step 1 — File a police complaint (NCR) at the nearest police station. Get a copy — this is needed for the claim.
Step 2 — Take clear photos of the damage from multiple angles before any repairs.

Step 3 — Call our claims number and register the claim. The claim number will be sent to you by SMS.
Step 4 — Take the car to an authorised repair centre in our network. Repairs at network garages are cashless — you pay nothing upfront.

If you have a Zero Depreciation add-on, you will get the full cost of parts replaced without any deduction. Shall I check whether your current policy includes Zero Dep?"

Life Insurance Questions

Q: "I missed 2 premium payments. Is my policy cancelled?"

"No, your policy is not cancelled immediately. After missing 2 premiums, your policy enters a Grace Period.

For monthly premium payment policies, the grace period is 15 days from the due date. If you pay within the grace period, the policy continues as if nothing happened — no penalty, no lapse.

If you do not pay within the grace period, the policy lapses. A lapsed policy does not provide coverage. However, you can revive a lapsed policy within 5 years by paying all outstanding premiums plus interest.

Your policy payment is monthly, so your grace period is 15 days from the due date of the second missed premium. I strongly recommend paying immediately to avoid any gap in coverage. Shall I show you the payment options?"

[Source: Section 5 — Premium Payment, Clause 5.3 — Grace Period and Lapse, Page 22]

Q: "if I die in an accident will my family get extra money"

"Yes. Your policy includes an Accidental Death Benefit rider.

If death occurs due to an accident:

- Basic Sum Assured: ₹50,00,000 (from the base life cover)
- Accidental Death Benefit: An additional ₹25,00,000 (from the rider)
- Total payout to your family: ₹75,00,000

The accident must be a sudden, unexpected, external, visible event and death must occur within 90 days of the accident.

Exclusions from the accidental death benefit (the basic sum assured is still paid in these cases):

- Death while under the influence of alcohol or drugs
- Self-inflicted injuries
- Participation in undeclared hazardous sports

Your nominee is currently listed as your wife. Do you want to verify the nominee details or update them?"

[Source: Section 6 — Riders, Clause 6.1 — Accidental Death Benefit, Page 29]

Network Hospital Questions

Q: "which hospitals near HSR layout bangalore are cashless for my plan?"

"Here are the cashless network hospitals near HSR Layout, Bengaluru that accept your Star Health Comprehensive plan:

1. **Manipal Hospital, HAL Airport Road** — 4.2 km away
   All specializations. Pre-authorization: 2 hours for planned, immediate for emergency.
   Insurance desk: 080-25023456
2. **Columbia Asia Hospital, Hebbal** — 6.8 km away
   Cardiology, Orthopaedics, General Surgery.
   Insurance desk: 080-61226122
3. **Sparsh Hospital, Infantry Road** — 7.1 km away
   Orthopaedics specialist. Strong for joint replacements.
   Insurance desk: 080-44660101

For emergency admission at any of these hospitals, go directly to the emergency ward and show your Star Health e-card. They will contact us directly. You do not need pre-authorization in emergencies.

For planned procedures, call the hospital's insurance desk at least 48 hours in advance to get cashless pre-authorization."

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15. Beyond Just Answering — What Else the Chatbot Can Do

Most people think a chatbot just answers questions. But a well-designed RAG chatbot does much more. It does not wait for the user to ask the perfect question — it proactively adds value at every interaction.

Feature 1 — Suggested Next Questions

After every answer, the chatbot suggests 3 to 4 related questions the user might want to ask next. This helps customers who do not know exactly what to ask, and surfaces benefits they did not know existed.

Example:

Customer asks: "Is my cataract surgery covered?"

Chatbot answers the question, then adds:

You might also want to know:

- *"How do I get cashless approval for cataract surgery?"*
- *"Which hospitals near me do cataract surgeries cashlessly?"*
- *"What documents do I need for the claim after surgery?"*
- *"Am I eligible for a free eye check-up under my plan?"*

The last question is the most important one. The customer may not have known they have a free annual eye check-up benefit. The chatbot surfaces it proactively.

Feature 2 — Proactive Benefit Discovery

The chatbot knows the customer's plan, sum insured, add-ons, and usage so far this year. It uses this to proactively tell customers about benefits they have not used.

Example:

Customer asks about a routine query. Chatbot answers, then adds:

"A quick note: You have ₹5,000 in unused OPD benefit remaining this policy year, which ends on 31 March 2025. This can be used for doctor consultations, medicines, diagnostic tests, and dental check-ups at empanelled clinics. Would you like to see the list of empanelled OPD clinics near you?"

The customer gets money they were entitled to but did not know about. The company earns trust and loyalty.

Feature 3 — Personalised Offers and Upgrade Suggestions

When the chatbot understands what a customer is asking about, it can recommend relevant products or upgrades at exactly the right moment.

Example 1 — Upgrade suggestion based on a complaint:

Customer: "Is my room rent limit ₹3,000 per day? That seems low."

Chatbot: "Yes, under your current Basic plan, room rent is covered up to ₹3,000 per day. For most hospitals in Tier 1 cities, this may require you to pay the difference for a standard room.

Your plan has an upgrade option — the Comprehensive Plus plan removes the room rent limit entirely. The additional premium is approximately ₹2,800 per year. Would you like me to generate a comparison?"

Example 2 — Add-on recommendation at the right moment:

Customer: "My car was scratched. I had to pay ₹8,000 for painting because of depreciation deduction."

Chatbot: "For car painting, standard depreciation on rubber and plastic parts is 50% for cars above 5 years. That is why you paid ₹8,000 of the total ₹16,000 bill.

Your current policy does not include the Zero Depreciation add-on. If you add it at renewal — which is coming up in 2 months — you would pay only the deductible (₹1,000) instead of ₹8,000 for any future own damage claim. The Zero Dep add-on costs approximately ₹1,400 per year for your car.

Over one claim, it pays for itself many times over. Want me to add this to your renewal preferences?"

Example 3 — Proactive renewal message with benefit review:

A week before renewal, the chatbot proactively messages:

"Your health policy renews on 15 April 2025. This year, you used ₹82,000 of your ₹5 lakh cover for your knee surgery. Your no-claim bonus of 10% will be paused this year, but it will be back next year if you do not make a claim.

Based on your usage this year, you may want to consider increasing your sum insured to ₹10 lakhs. The additional premium is approximately ₹1,900 per year for your age group.

You also have a wellness reward of 500 points from your health check-up last June that can be redeemed against your renewal premium. Shall I apply it?"

Feature 4 — Claims Estimation Before Filing

Before a customer files a claim, the chatbot gives them a realistic estimate of what they will receive — so there are no surprises.

Example:

Customer: "I am being discharged tomorrow. My total bill is ₹1,40,000. How much will I get?"

Chatbot: "Let me calculate your expected reimbursement based on your policy:

Item	Hospital Bill	Your Policy Covers	You Pay
Room rent (5 nights at ₹6,000/day)	₹30,000	₹15,000 (limit ₹3,000/day)	₹15,000
Surgery charges	₹80,000	₹80,000	₹0
Anaesthesia	₹12,000	₹12,000	₹0

Item	Hospital Bill	Your Policy Covers	You Pay
Medicines during stay	₹8,000	₹8,000	₹0
Diagnostic tests	₹10,000	₹10,000	₹0
Total	**₹1,40,000**	**₹1,25,000**	**₹15,000**

Your main out-of-pocket cost is the excess room rent — your policy covers ₹3,000 per day but the hospital charged ₹6,000. Everything else is fully covered.

Would you like me to tell you what documents you need to collect from the hospital at discharge?"

This transparency builds enormous trust. The customer knows exactly what to expect before they leave the hospital.

Feature 5 — Document Checklist Generator

Rather than just saying "you need documents", the chatbot generates a specific, personalised checklist based on the claim type and hospital.

Example:

Customer: "I am filing a claim for my mother's hip replacement surgery at Apollo Hospitals, Chennai. What documents do I need?"

Chatbot: "Here is your complete document checklist for this claim:

From Apollo Hospitals:

- [ ] Original discharge summary with diagnosis (ICD code) and treatment details
- [ ] Final itemised hospital bill with Apollo letterhead
- [ ] All payment receipts
- [ ] Operation notes signed by the surgeon
- [ ] Implant sticker or invoice — important for hip replacement, always requested
- [ ] Pre-operative reports: blood tests, X-rays, MRI scan

From the doctor:

- [ ] Referral letter or consultation notes recommending the surgery

- [ ] Post-operative follow-up prescription

From you:

- [ ] Duly filled claim form
- [ ] Copy of your mother's Aadhaar card
- [ ] Your policy card copy
- [ ] Bank account details for reimbursement

One thing specific to hip replacement claims:
Star Health requires the implant invoice separately from the hospital bill. Make sure you ask Apollo's billing department to give you the implant cost as a separate line item. This is commonly missed and delays claims.

Shall I also tell you the expected timeline for this claim?"

Feature 6 — Appointment and Service Integration

For insurers that have connected their backend systems, the chatbot can go beyond information and actually complete actions.

What the chatbot can do:

- Book a free health check-up at an empanelled diagnostic centre — right from the chat
- Schedule a field executive visit to collect claim documents at home
- Raise a pre-authorization request for a planned surgery
- Update nominee details after verification
- Request a duplicate policy document by email
- Initiate a portability request if the customer wants to switch insurers
- Renew the policy and generate a payment link

Example:

Customer: "I want to book the free health check-up. I am in Pune."

Chatbot: "Your policy includes a free comprehensive health check-up once a year — you have not used it this year yet.

Here are two available slots at SRL Diagnostics, Baner, Pune (the nearest empanelled centre):

- Wednesday 19 March 2025, 7:00 AM
- Thursday 20 March 2025, 7:30 AM

The check-up covers: CBC, blood sugar, lipid profile, thyroid, urine routine, chest X-ray, and ECG.

Which slot works for you? I will send the confirmation and address to your registered mobile number."

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16. What Makes This Project Production-Grade?

A student project stops at getting the chatbot to answer questions correctly in a notebook. A production project handles everything that happens in the real world.

Multi-Tenant Isolation: Star Health's policy documents must never be accessible to Bajaj Allianz users. Each company gets a completely separate space in the vector database. Even if they share the same server infrastructure, the data walls are absolute.

Hallucination Prevention: The system never answers from general knowledge. If the answer is not in the uploaded documents, it says so. After generating an answer, a second AI model checks whether the answer is actually supported by what was retrieved. If not, confidence is marked low and the customer is told to speak to an agent.

Audit Trail: Every single interaction is logged — who asked, what they asked, which documents were searched, what chunks were retrieved, what the answer was, and what the confidence score was. This is not optional in insurance — IRDAI requires it.

Response Caching: The same question gets asked thousands of times a day. "What is the waiting period for pre-existing conditions?" does not need a fresh AI call every time. The first time it is asked, the answer is cached. Subsequent identical or similar questions are served from cache instantly. This reduces AI API cost by 40 to 60%.

Guardrails: The system blocks questions containing personal information like Aadhaar numbers. It blocks attempts to manipulate the AI with hidden instructions. It blocks questions unrelated to insurance. Every output is checked for harmful content before being shown to the customer.

Multilingual Support: A policyholder in Tamil Nadu asks in Tamil. A customer in Gujarat asks in Gujarati. The system handles queries in any Indian language and responds in the same language the question was asked.

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17. Project Deliverables

At the end of this project, the following are delivered:

Working Chatbot Application: A web-based interface where users can upload policy documents and ask questions. Built with Streamlit for the demo and React for production deployment.

REST API: A documented API that the insurance company's existing systems can call. Their mobile app, website, and call centre tool can all integrate with the same chatbot backend.

Document Ingestion Pipeline: An automated system that processes new policy documents as soon as they are uploaded — no manual steps needed.

Admin Dashboard: A dashboard where the insurance company's team can see which questions are being asked most often, which answers have low confidence, and which documents are being queried.

Audit Log System: A searchable log of all interactions — required for IRDAI compliance.

AWS Deployment: Everything deployed on AWS — the application, the database, the AI models, the monitoring. Includes CI/CD pipeline for automated deployments.

LLMOps Monitoring: LangSmith traces every AI call. CloudWatch monitors system health. Alerts fire when hallucination rate or response time crosses acceptable thresholds.

Documentation: Complete technical documentation for the engineering team and a non-technical guide for the business team.

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